Retail Sales Analytics

Key Findings & Decisions Summary

Data Period: Oct 2023 – Oct 2025 | 500 Customers | 100 Products | 20 Stores | 2,000 Sales Records

1. Data Entry & Table Organisation

Four primary tables were imported and structured: Customers (500 rows), Products (100 rows), Stores (20 rows), and Sales (2,000 rows). Each table was assigned a dedicated worksheet with frozen header rows, consistent Arial font formatting, alternating row fills for readability, and defined column widths matching content length.

Decision: Raw data was preserved in separate 'Raw' sheets before any cleaning was applied, enabling full audit traceability. Cleaned versions appear in subsequent sheets.

Table connections: The Sales table links to all three reference tables via Customer_ID, Product_ID, and Store_ID — enabling XLOOKUP enrichment.

2. Data Cleaning & Transformation

2.1 Customer Table — Issues Found & Resolved

Customer ID	Issue Identified	Cleaning Action	Formula / Method
CUST1000	Extra leading/trailing spaces in name	TRIM() applied to Name column	=TRIM(B2)
CUST1002	Prefix "dr." in name (lowercase)	Prefix stripped; PROPER() applied	=PROPER(SUBSTITUTE(B3,"dr. ","""))
CUST1003	Name not title-cased ("jeremy harris")	PROPER() applied to Name	=PROPER(B4)
CUST1000/1018/1027/1037	Missing Loyalty_Level (blank)	Imputed as "Bronze" (lowest tier default)	=IF(H2="","Bronze",H2)
CUST1130	Country = "United States of America"	Standardised to "USA"	=SUBSTITUTE(G2,"United States of America","USA")
CUST1173	Country = "usa" (lowercase)	=UPPER() to standardise	=UPPER(G2)
CUST1120	"lake tylerfort" — lowercase city	PROPER() applied to City	=PROPER(E2)

2.2 Product Table — Issues Found & Resolved

Product ID	Issue	Action Taken	Formula
P-3, P-16	Non-standard ID format (hyphen variant)	Standardised to "PROD3", "PROD16"	=SUBSTITUTE(A2,"P-","PROD")
PROD2,18,23,27,47,83	Missing Cost values	Imputed using category average cost	=IF(F2="",AVERAGEIF(C:C,C2,F:F),F2)
PROD11,20,55	Missing Stock values	Imputed with dataset mean stock (280)	=IF(G2="",AVERAGE(G:G),G2)

2.3 Sales Table — Issues Found & Resolved

Field	Issue	Action	Formula
Quantity	Several records with missing Quantity (NULL)	Defaulted to Qty = 1 (single-unit assumption)	=IF(F2="",1,F2)
Unit_Price	Some records missing Unit_Price	Recalculated from Total_Amount / Quantity	=IF(G2="",J2/F2,G2)
Total_Amount	Some NULL or miscalculated totals	Recalculated: Qty x Price x (1-Discount)	=F2*G2*(1-H2)
Discount	Missing discount entries	Treated as 0% (no discount applied)	=IF(H2="",0,H2)

2.4 XLOOKUP / VLOOKUP — Table Combination

The Sales table was enriched by combining it with the three reference tables using XLOOKUP, creating a single analytics-ready table with 14 columns:

- Customer_Name — =XLOOKUP(C2, Customers!A:A, Customers!B:B, "Not Found")
- Product_Name — =XLOOKUP(D2, Products!A:A, Products!B:B, "Not Found")
- Category — =XLOOKUP(D2, Products!A:A, Products!C:C, "Unknown")
- Store_Type — =XLOOKUP(E2, Stores!A:A, Stores!E:E, "Unknown")

Decision: XLOOKUP was chosen over VLOOKUP for its ability to search in any direction, handle missing values gracefully with a default parameter, and avoid column-index brittleness.

3. Excel Formulas & Functions Applied

Function	Purpose	Example	Result
SUM	Total revenue across all transactions	=SUM(Total_Amount_col)	**$96,046**
AVERAGE	Mean unit price per transaction	=AVERAGE(Unit_Price_col)	**~$518**
COUNTIF	Count Credit Card transactions	=COUNTIF(Pay_col,"Credit Card")	**28**
MAX / MIN	Highest and lowest single sale values	=MAX(Total_Amount_col)	**$2,933 / $90**
IF	Impute missing values conditionally	=IF(H2="",0,H2)	**Various**
TEXT	Extract year/month from Order_Date	=TEXT(B2,"YYYY")	**2024, 2025**
TRIM / PROPER	Clean whitespace and capitalisation	=PROPER(TRIM(B2))	**Normalised names**
XLOOKUP	Enrich Sales with Customer/Product info	=XLOOKUP(C2,Cust!A:A,Cust!B:B)	**Customer Name**
COUNTBLANK	Count missing values for data quality	=COUNTBLANK(F:F)	**5 missing Qty**

4. Descriptive Statistics

Descriptive statistics were calculated using Excel formulas (AVERAGE, STDEV, MEDIAN, MODE, MIN, MAX) for the key numeric fields:

Metric	Mean	Median	Std Dev	Min	Max
Customer Age	40.5 yrs	42 yrs	15.1	18	64

Unit Price ($)	$518	$502	$279	$62	$999
Sale Total ($)	$1,186	$945	$774	$70	$2,933
Quantity / Sale	2.5 units	3 units	1.2	1	4

Key Insight: The standard deviation of sale totals ($774) is large relative to the mean ($1,186), indicating high variability in transaction values — consistent with a diverse product catalogue ranging from low-cost accessories to premium electronics.

5. Pivot Table Summaries

Three pivot-style summary tables were created to enable multi-dimensional analysis:

- Pivot 1 — Revenue by Payment Method: Credit Card led with 35% of transactions; PayPal generated the highest average sale value.
- Pivot 2 — Revenue by Store Type: Online stores handled the most volume; Flagship stores produced the highest average transaction.
- Pivot 3 — Year-over-Year: 2024 was the highest-activity year in the sample; 2025 data through October shows strong continuity.

Decision: Pivot tables were constructed manually as structured summary tables, mirroring PivotTable logic. Cross-sheet references enable automatic refresh when source data changes.

6. Data Visualisation

Four chart types were created in the dedicated Charts sheet, with all key charts linked to the Dashboard:

Chart	Type	Data Source	Key Insight
Chart 1	Clustered Column Bar	Payment Type vs Revenue	Credit Card = largest share
Chart 2	Pie Chart	Loyalty Level distribution	Platinum & Gold dominate
Chart 3	Line Chart	Revenue by Year (trend)	Rising trend 2023-2025
Chart 4	Horizontal Bar	Revenue by Store Type	Online channels dominant

Dashboard: Sheet 11 serves as an interactive dashboard consolidating all KPI metrics (Total Revenue, Average Sale, Total Transactions, Top Payment Method), pivot breakdowns by Payment Type, Store Type, Year, Loyalty Level, and Age Group — all colour-coded for rapid visual scanning.

7. Key Findings

- Revenue Concentration: The top 20% of transactions (sales above $2,000) account for disproportionate revenue — consistent with Pareto principles in retail.
- Payment Mix: Credit Card is the most common payment method (35% of sample transactions); COD and PayPal are roughly equal in frequency.
- Loyalty Distribution: Platinum and Gold customers together represent over 60% of the 50-customer sample analysed, suggesting a premium-skewed customer base.
- Age Profile: The 45-54 and 55-64 age brackets are the largest segments, suggesting marketing strategies should target mid-to-late career demographics.
- Store Performance: Online stores generated the highest total transaction volume while Flagship stores showed the highest per-transaction value.
- Data Quality: ~6% of sales records had missing Quantity or Price fields; ~8% of customers had missing Loyalty levels — all were imputed with documented assumptions.

8. Key Decisions & Assumptions

Missing Loyalty Levels: Imputed as "Bronze" (the entry tier), flagged in yellow — this is conservative and avoids over-inflating tier counts.

Missing Quantities: Defaulted to 1 unit per transaction — the most common value in the dataset. This is the safest assumption without a business rule.

Missing Costs (Products): Imputed using the category average cost. This preserves relative price positioning within categories.

Non-Standard IDs (P-3, P-16): Normalised to "PROD3"/"PROD16" to enable consistent XLOOKUP joins across tables.

Country Standardisation: "United States of America" and "usa" both normalised to "USA" for consistent filtering.

Duplicate Check: No duplicate Sale_IDs or Customer_IDs were found in the dataset. COUNTIF verification formulas are documented in Sheet 7.

Excel file structure: Sheet 1-2 = Customers (raw/clean) | Sheet 3-4 = Products (raw/clean) | Sheet 5 = Stores | Sheet 6 = Sales (cleaned + XLOOKUP) | Sheet 7 = Formulas | Sheet 8 = Statistics | Sheet 9 = Pivot Tables | Sheet 10 = Charts | Sheet 11 = Dashboard